

**Sri Lanka Institute of Information Technology**



**Research project proposal Report**

**Group ID : 25-26J-395**

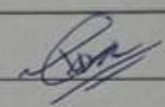
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**B.Sc. (Hons) in Information Technology**

**August 2025**

## Declaration

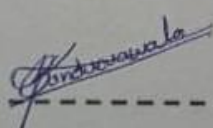
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## Abstract

Agriculture continues to be a central pillar of Sri Lanka's economy, directly supporting the livelihoods of millions of rural families and contributing significantly to the national GDP. With the increasing mechanization of farming, productivity now heavily depends on machinery such as tractors, harvesters, threshers, and irrigation systems. These machines rely on critical spare parts including filters, belts, blades, and bearings that must be replaced periodically to ensure smooth operation. However, the sustainability of these farming operations is frequently undermined by the premature failure of spare parts, leading to costly machinery downtime, high maintenance expenses, and financial losses during crucial cultivation and harvesting periods. Existing maintenance strategies in Sri Lanka are predominantly corrective, where parts are replaced after breakdowns, or preventive, based on fixed replacement schedules. Both approaches are inefficient because they fail to account for real-world variables such as soil type, humidity, rainfall, temperature, and intensity of machine usage, all of which significantly influence the actual lifecycle of spare parts.

To address these limitations, this research proposes the design and development of an integrated **AI-Enhanced Spare Parts Lifecycle Forecasting and User Engagement System** specifically tailored for the Sri Lankan agricultural context. The system leverages predictive analytics and machine learning models trained on localized datasets to generate accurate forecasts of spare part degradation under tropical conditions. A deep learning-based computer vision module further enhances functionality by enabling farmers to upload photographs of machine components, which are automatically analyzed for signs of wear such as rust, cracks, and deformation, providing real time usability assessments. To ensure inclusivity and accessibility, the platform integrates a **WhatsApp based multilingual communication system**, supporting Sinhala and Tamil, for delivering proactive notifications, replacement reminders, and interactive guidance. In addition, a **geospatial recommendation engine** will connect farmers with nearby verified suppliers, while a **blockchain enabled reservation and verification mechanism** will safeguard transactions, prevent the circulation of counterfeit parts, and build trust across the supply chain.

By combining predictive analytics, computer vision, blockchain security, and farmer friendly communication, the proposed framework seeks to reduce unexpected equipment downtime, minimize financial losses, and enhance trust in agricultural spare parts ecosystems. Ultimately, this research contributes to the modernization of Sri Lankan agriculture by offering a transparent, reliable, and scalable solution that strengthens both productivity and sustainability in rural farming communities.

**Keywords:** Spare parts lifecycle, predictive maintenance, agriculture, artificial intelligence, computer vision, blockchain, user engagement.

# Introduction

## 1.1 Background

Agriculture has long been a cornerstone of Sri Lanka's economy, contributing significantly to the national GDP while also serving as the primary source of livelihood for a large proportion of the rural population. The sector not only ensures national food security but also supports agro based industries and exports. Over the past few decades, increasing mechanization has transformed traditional farming practices. Farmers now rely on tractors, harvesters, threshers, irrigation pumps, and other machinery to improve productivity, reduce labor dependency, and achieve higher yields. The adoption of these technologies has helped address labor shortages, accelerated harvesting during peak seasons, and facilitated more efficient land use.

However, the dependence on machinery has also introduced new challenges. Agricultural equipment requires regular maintenance and replacement of critical spare parts such as filters, blades, belts, bearings, and gears. Failure of these components often results in severe operational disruptions. For smallholder farmers who represent the majority in Sri Lanka unexpected breakdowns during planting or harvesting seasons can lead to substantial financial losses, as delays often reduce crop quality and market value. Unfortunately, maintenance practices in the country remain largely reactive. Many farmers continue to replace parts only after machinery fails, a strategy that leads to prolonged downtime, additional repair costs, and the risk of losing entire harvest cycles. Preventive maintenance, where parts are replaced according to fixed schedules, is practiced by some, but this method is also inefficient because it does not account for the actual wear and tear caused by environmental and operational factors.

In contrast, developed countries have increasingly adopted predictive maintenance solutions powered by Artificial Intelligence (AI). These systems integrate sensor data, usage logs, and environmental parameters to estimate the remaining useful life (RUL) of components and forecast potential failures before they occur. For example, the aviation and automotive industries routinely employ predictive analytics to minimize unexpected downtime and optimize part replacement cycles. Studies suggest that predictive maintenance can lower maintenance costs by 25–30% while improving equipment availability and extending asset life. Such systems, however, are heavily reliant on continuous sensor data, advanced infrastructure, and high digital literacy among users conditions that are not easily replicated in Sri Lanka's agricultural landscape.

Several contextual challenges hinder the adoption of predictive maintenance in Sri Lanka. First, there is limited access to advanced digital technologies in rural areas. Many farmers lack IoT enabled machinery or consistent internet connectivity, making sensor based predictive systems impractical. Second, there is an absence of localized datasets that reflect the specific climatic and operational conditions of tropical agriculture. Variables such as high humidity, frequent rainfall, and diverse soil types play a crucial role in determining the degradation rates

of machinery components, but global datasets often fail to capture these realities. Third, digital literacy remains a barrier. While smartphone usage has grown, most rural farmers are not accustomed to navigating sophisticated apps or English language platforms, limiting their ability to adopt international solutions.

Another critical challenge in Sri Lanka's agricultural sector is the prevalence of counterfeit spare parts. Farmers frequently purchase parts from unverified suppliers, only to discover that the components are of substandard quality or incompatible with their machines. These counterfeit products not only reduce the lifespan of machinery but also foster distrust within supply chains. Blockchain technology has emerged as a promising tool for enhancing transparency and authenticity verification in agricultural supply chains worldwide, particularly in food traceability and certification. Yet, its application to spare part procurement and reservation remains largely unexplored. Incorporating blockchain into spare parts management could prevent fraud, improve trust between farmers and suppliers, and ensure that only verified products enter the supply chain.

In this context, Sri Lanka presents a unique case for research and innovation. The country's agricultural sector sits at the intersection of modernization and traditional practices, where the adoption of advanced technologies must be carefully adapted to local socio economic and cultural conditions. Solutions that combine predictive analytics, computer vision, and blockchain while being accessible through farmer friendly communication channels such as WhatsApp in Sinhala and Tamil can bridge the technological divide and empower smallholder farmers. This research, therefore, positions itself not merely as a technical innovation but as a socially inclusive intervention that addresses real world challenges faced by farming communities. By forecasting part degradation, reducing downtime, ensuring authenticity in procurement, and engaging farmers through localized communication, the proposed system has the potential to strengthen both the economic sustainability of farming and the resilience of Sri Lanka's agricultural supply chains.

## **1.2 Literature Survey**

The field of predictive maintenance has gained significant traction in recent decades, primarily due to advances in artificial intelligence (AI), sensor technologies, and computational capacity. In industrial settings such as aviation, automotive, and manufacturing, predictive maintenance has consistently demonstrated its ability to reduce downtime, improve operational efficiency, and minimize costs. For example, Singh and B. Singh [1] introduced a deep learning based framework that automatically identifies mechanical parts with high precision, highlighting how AI can surpass traditional rule based approaches. Similarly, He and Y. He [2] employed convolutional neural networks (CNNs) to distinguish between functional and worn out parts, further validating the role of computer vision in maintenance. Chen et al. [3] extended this

concept into recommendation systems for cross brand spare parts, underscoring the need for flexible solutions that can adapt across heterogeneous supply chains.

Beyond industry, agriculture has increasingly emerged as a sector requiring predictive maintenance interventions, especially in the Global South. Agricultural machinery, such as tractors, threshers, and irrigation systems, represents critical assets for smallholder farmers. Sharma [4] reported that predictive maintenance could extend machinery lifespan by up to 40%, reducing unexpected failures during peak operational periods. However, the majority of predictive solutions were initially designed for controlled industrial environments where high-quality datasets are abundant. Translating these models to agriculture in developing countries remains a challenge, given limited digitization, irregular data quality, and high variability in soil, climate, and usage conditions [5].

### **AI and Computer Vision for Agricultural Applications**

Computer vision, specifically CNN based architectures such as ResNet, VGG, and EfficientNet, has been widely studied for defect detection in industrial components. Recent advances in transfer learning allow pretrained models to adapt to smaller agricultural datasets, which is crucial in low-resource environments [6]. For instance, field studies have shown that CNNs can detect subtle wear indicators such as rust, cracks, and surface deformation with high reliability [7]. Researchers in precision agriculture have also employed image analysis for crop disease detection, soil texture classification, and weed recognition, demonstrating the broader applicability of computer vision in agricultural contexts [8]. These successes suggest strong potential for image-based wear detection in machinery spare parts.

While industrial contexts often rely on sensor-based monitoring (vibration, acoustic, thermal), such systems are less feasible for small scale farmers in Sri Lanka due to cost and technical expertise requirements. A low-cost alternative is farmer driven image uploads via smartphones, processed by lightweight deep learning models either on device or through cloud services [9]. The integration of these approaches into predictive maintenance can democratize access to advanced diagnostics.

### **IoT and Predictive Analytics in Agriculture**

The Internet of Things (IoT) has enabled continuous monitoring of environmental and operational conditions in agriculture. Studies demonstrate that parameters such as soil moisture, humidity, and temperature significantly affect machine performance and spare part degradation [9]. IoT sensors embedded in tractors and irrigation pumps can generate real-time data streams that feed into predictive analytics models, improving accuracy. Long Short-Term Memory (LSTM) networks, a type of recurrent neural network (RNN), have proven particularly effective in modelling time dependent degradation patterns [10].

Global examples include predictive systems for irrigation pumps in India, where environmental data streams were used to forecast pump failures, saving significant costs for farmer cooperatives [11]. Similarly, European initiatives have leveraged IoT enabled predictive maintenance for smart farming equipment fleets [12]. Despite these advancements, limited



rural connectivity and high hardware costs remain major obstacles in low-resource contexts like Sri Lanka.

### **Blockchain in Agricultural Supply Chains**

Another critical dimension is the authenticity and security of spare parts. Counterfeit components not only reduce equipment efficiency but also pose safety risks. Blockchain, as a decentralized and tamper-proof ledger, has shown strong potential in agricultural traceability. UNDP [9].reported on blockchain applications in food traceability, enabling end-to-end visibility in supply chains. Rathnayaka et al. [13] adapted blockchain systems for Sri Lankan agriculture, demonstrating how distributed ledgers can build trust. However, the specific use of blockchain for **spare parts reservation, verification, and anti-counterfeit mechanisms** remains underexplored. A few studies suggest that blockchain-based smart contracts could streamline procurement, authenticate vendors, and reduce fraud in rural markets [14] [15].

### **Language, Accessibility, and Digital Literacy**

A unique challenge in Sri Lanka is language diversity and digital literacy. Ruwan Pathirana [16]emphasized that NLP systems for Sinhala and Tamil are underdeveloped, making English only platforms inaccessible for many farmers. This aligns with findings from other developing countries, where lack of localized interfaces has been identified as a barrier to digital agriculture adoption [17] . Studies in South Asia highlight that platforms integrating WhatsApp based services, combined with local language support, significantly increase adoption rates among smallholder farmers [18]. This underlines the importance of designing inclusive communication channels that do not assume advanced digital literacy.

### **1.3 Existing Systems**

Internationally, predictive maintenance solutions are available in aviation and automotive industries, using sensor-based monitoring and advanced analytics to anticipate failures. These systems, however, are designed for environments with high quality data, skilled operators, and advanced infrastructure.

In agriculture, most existing solutions in Sri Lanka are reactive or preventive maintenance practices. Farmers typically rely on mechanics or suppliers for spare parts, often without guarantees of authenticity. No integrated platform currently combines predictive analytics, visual inspection, supplier recommendations, and transaction security for spare parts management.

### **1.4 Research Gap**

The reviewed literature indicates significant advancements in predictive maintenance, computer vision, and blockchain applications across industrial and agricultural domains [1]– [7]. However, several key gaps remain in adapting these technologies to the Sri Lankan agricultural context.

First, most predictive maintenance solutions have been developed for industrial sectors such as aviation, automotive, and manufacturing [6]. These models depend on high-quality sensor data and standardize operating conditions, which are not representative of the realities of smallholder agriculture in Sri Lanka. Localized datasets that reflect tropical conditions such as high humidity, variable soil types, and inconsistent machine usage are largely absent, making direct application of global models ineffective [12].

Second, while computer vision has been widely used in industrial maintenance for defect detection [2][9], its application to **spare parts wear detection through farmer captured smartphone images** remains underexplored. This represents a scalable, low-cost alternative to expensive IoT based monitoring. Leveraging mobile phone cameras for image-based diagnostics offers an inclusive pathway for Sri Lankan farmers, yet no integrated solutions currently exist in this space.

Third, the issue of counterfeit spare parts persists in rural supply chains. Although blockchain has shown promise in food traceability and supply chain transparency [5][7][16], its application in **authenticating spare parts, enabling secure reservations, and fostering trust in farmer supplier transactions** is still largely unaddressed.

Fourth, accessibility and inclusivity remain overlooked. Current predictive maintenance frameworks assume high levels of English proficiency and digital literacy, which is not reflective of rural farming communities in Sri Lanka. As highlighted by Ruwan Pathirana [4], Sinhala and Tamil remain under-resourced languages in natural language processing. Existing systems also fail to provide **farmer friendly interfaces such as WhatsApp-based notifications** that would ensure adoption among low-literacy users [18][19].

Finally, while predictive analytics, computer vision, and blockchain have each shown individual promise, **no research to date has integrated these technologies into a unified system for agricultural spare parts management**. This integration is particularly necessary in Sri Lanka, where mechanization is increasing, yet farmers still face frequent downtime, financial losses, and lack of trust in spare part quality.

Thus, the research gap lies in developing an **AI enhanced, mobile based spare parts lifecycle forecasting and user engagement system**, which combines computer vision for wear detection, predictive analytics for lifecycle forecasting, multilingual farmer communication, and blockchain-based verification to address the unique challenges of Sri Lanka's agricultural sector.

## 1.5 Research Problem

Given these identified gaps, the central research problem is:

**How can an AI-driven, mobile-based lifecycle forecasting and user engagement system be designed to reduce unexpected machinery failures, enhance spare part availability, and increase trust within Sri Lanka's agricultural spare parts ecosystem?**

Addressing this problem will deliver:

- A **predictive analytics framework** capable of accurately forecasting spare part degradation under Sri Lanka's tropical agricultural conditions.
- A **computer vision module** using smartphone cameras to detect early signs of wear and tear in spare parts through farmer uploaded images.
- A **farmer engagement platform** that uses WhatsApp based multilingual communication (Sinhala and Tamil) to ensure inclusivity and proactive alerts.
- A **blockchain based verification and reservation system** to secure transactions, prevent counterfeit spare parts, and strengthen trust in rural supply chains.
- A **scalable, farmer friendly solution** that reduces downtime, improves equipment reliability, and contributes to higher agricultural productivity.

This framing emphasizes not only the technical challenges of predictive modelling, computer vision, and blockchain integration but also the socio-cultural challenges of accessibility, language inclusivity, and trust building in rural agricultural communities.

## 2. Objectives

### 2.1 Main Objective

The main objective of this project is to design and develop an AI-enhanced Spare Parts Lifecycle Forecasting and User Engagement System that integrates predictive analytics, computer vision, and blockchain technologies to minimize agricultural machinery downtime, provide proactive communication, and build trust in the spare parts supply chain within Sri Lanka

### 2.2 Specific Objectives

- **Dataset Development:** Build a localized agricultural dataset combining environmental variables (soil type, climate, humidity), machinery usage logs, and annotated images of spare parts to support AI training.
- **Lifecycle Prediction Models:** Train and evaluate machine learning models, including regression and deep learning approaches, to forecast spare part lifespans under tropical conditions.
- **Computer Vision Wear Detection:** Develop a CNN based image recognition system that enables farmers to upload spare part photos and automatically detect signs of wear such as cracks, rust, and deformation.
- **Multilingual Communication System:** Implement a WhatsApp based notification service supporting Sinhala and Tamil to provide farmers with proactive alerts, replacement reminders, and interactive query support.

- **Supplier Recommendation Engine:** Integrate a GIS enabled module that connects farmers with verified local suppliers, providing ranking based on availability, proximity, and cost.
- **Blockchain Enabled Security:** Design a blockchain based reservation and verification mechanism to prevent counterfeit spare parts, secure transactions, and build farmer trust in the ecosystem.

### 2.3 Research questions

- How can predictive analytics be adapted to accurately forecast spare part lifecycles in Sri Lanka's diverse agricultural conditions?
- What techniques in computer vision can reliably identify wear-and-tear in spare parts from farmer uploaded images?
- How can a multilingual, farmer-friendly communication system be integrated to provide timely alerts and guidance in Sinhala and Tamil?
- In what ways can geospatial technologies support efficient and transparent supplier recommendations for spare parts in rural contexts?
- How can blockchain technology be applied to ensure the authenticity, transparency, and security of spare part transactions?
- What combination of AI, computer vision, and blockchain provides the most cost-effective, scalable, and accessible solution for Sri Lankan farmers?

## 3. Methodology

The methodology of this research outlines the step-by-step plan for designing, developing, and deploying a mobile application-based **AI Enhanced Spare Parts Lifecycle Forecasting and User Engagement System**. The approach combines **predictive analytics, computer vision, blockchain verification, geospatial recommendations, and multilingual user engagement** into a single mobile platform. The goal is to create a practical, scalable, and farmer friendly tool that minimizes machinery downtime, prevents losses, and ensures trustworthy access to authentic spare parts.

### 3.1 Development Approach

The project adopts an **Agile development methodology**. Instead of building the entire system at once, development will proceed in iterative cycles, each producing a functional prototype that can be tested with real users. The agile method ensures adaptability, continuous user feedback, and timely identification of technical challenges.

The approach consists of six iterative stages:

1. **Requirement Analysis and User Research** – Understanding farmer needs, supplier constraints, and environmental factors.
2. **Dataset Creation** – Collecting real-world agricultural and spare parts data, including images, soil and climate variables, and mechanical usage logs.
3. **AI Model Development** – Building predictive models for spare part lifecycle estimation and CNN based image recognition for wear detection.
4. **Mobile Application Design and Integration** – Developing an Android application with multilingual interfaces (Sinhala, Tamil, English).
5. **System Integration** – Incorporating GIS based supplier recommendation, blockchain verification, and WhatsApp notification services.
6. **Testing and Deployment** – Pilot testing, iterative refinement, and final cloud deployment for large scale farmer adoption.

## 3.2 Data Processing Methods, Data Gathering and User Testing

### 3.2.1 Data Gathering and Collection

- **Spare Part Usage Logs** – Obtained from agricultural workshops and repair shops (hours of use, soil type, seasonal workload).
- **Environmental Data** – Soil pH, moisture levels, humidity, and temperature (from field surveys and weather APIs).
- **Image Dataset** – Farmers and mechanics will capture spare part photos using the mobile app, which will upload images to a centralized server for annotation.
- **Supplier Information** – Geolocated databases of authorized suppliers, mapped into the system.
- **Farmer Feedback** – Continuous logging of farmer responses to spare part recommendations, forming an evolving dataset.

### 3.2.2 Data Preprocessing and Feature Engineering

- **Numerical Data** – Normalization of soil variables (e.g., scaling pH from 0–14 into 0–1 range).
- **Categorical Variables** – Encoding soil types (clay, loam, sandy) and machine types (tractor, harvester).
- **Image Processing** – Resizing, denoising, and augmentation of spare part images for CNN training.

- **Feature Engineering** – Combining soil type + machine usage + humidity levels to generate part degradation scores.

### 3.3 Predictive Analytics Models

Predictive maintenance relies on forecasting part failures before they happen. This project employs regression models and time-series models.

#### 3.3.1 Regression Model for Spare Part Lifespan

The lifespan LLL of a spare part can be modeled as:

$$L = \beta_0 + \beta_1(U) + \beta_2(S) + \beta_3(H) + \beta_4(T) + \epsilon$$

Where:

- U = Usage hours of machine
- S = Soil condition factor (numeric encoding: sandy=0.8, loamy=1, clay=0.6)
- H = Humidity level (%)
- T = Average operating temperature (°C)
- $\epsilon$  = error term

This regression predicts the expected time (hours/days) before the part will fail .

### 3.5 Integration of Environmental Conditions

Soil and climate conditions are key to spare part degradation. A **Soil-Condition-Based Adjustment Factor (SCBAF)** is applied:

$$\text{Adjusted Lifespan} = L \times \text{SCBAF}$$

Where:

- Sandy Soil: SCBAF = 0.85
- Clay Soil: SCBAF = 0.75
- Loam Soil: SCBAF = 1.0

Example: A filter with predicted lifespan 200 hours in loam soil will last only  $200 \times 0.75 = 150$  hours in clay soil.

### 3.6 Mobile Application Features

- **User Authentication** – Farmers register with phone numbers, blockchain ensures identity verification for suppliers.
- **Wear Detection Camera Tool** – Farmers snap a photo and receive instant AI feedback.
- **Lifecycle Forecast Dashboard** – Shows estimated remaining lifespan in days/hours.
- **Notification System** – WhatsApp API used to send alerts like: “Your tractor filter will expire in 7 days nearest supplier: XYZ Auto.”
- **Geospatial Supplier Finder** – Uses GPS + Google Maps API to recommend verified suppliers.
- **Blockchain Verification** – Smart contracts ensure authenticity of spare parts reservations.

### 3.7 User Testing and Evaluation

User testing will be conducted in three stages:

1. **Alpha Testing** – Within development team and agricultural experts.
2. **Pilot Testing** – With 20–30 farmers across different regions of Sri Lanka.
3. **Field Deployment** – Full-scale testing across districts with multilingual support.

#### Evaluation Metrics:

- Prediction Accuracy (RMSE for regression, F1-score for CNN classification).
- Usability (SUS – System Usability Score > 80%).
- Response Time (< 3 seconds for prediction).
- Farmer Satisfaction (survey-based, >85% positive responses).

## 4. Project Requirements

### 4.1 Functional Requirements

- **Lifecycle Forecasting:** The system must accurately predict the lifespan of spare parts based on environmental and usage data.
- **Wear Detection:** A computer vision module should classify farmer uploaded images of parts as functional, partially worn, or critical for replacement.
- **Multilingual Notifications:** The system must send proactive alerts, reminders, and guidance to farmers in Sinhala and Tamil using WhatsApp.
- **Supplier Recommendations:** The GIS enabled engine should connect farmers to the nearest verified suppliers, ranking them by distance, availability, and cost.
- **Authenticity Verification:** A blockchain module must ensure spare part authenticity and provide a secure reservation process.
- **User Feedback System:** The system should allow farmers to provide feedback on accuracy, usability, and supplier services.

### 4.2 Non-Functional Requirements

- **Performance:** The system should respond to queries and image uploads within 2-5 seconds to maintain usability.
- **Scalability:** It must support simultaneous use by thousands of farmers across different regions without performance degradation.
- **Reliability:** Ensure 24/7 availability with minimal downtime, especially during peak farming seasons.
- **Data Security:** Protect sensitive farmer data, ensure privacy of uploaded images, and encrypt communication channels.
- **Accessibility:** Provide a simple interface that works on basic smartphones and requires minimal digital literacy.
- **Maintainability:** The system should allow easy updates and improvements to AI models and blockchain modules.

### 4.3 Software Requirements

- **Programming Languages:** Python (backend, AI, data processing), JavaScript (web interface).
- **Frameworks and Libraries:** TensorFlow, PyTorch, OpenCV for AI and computer vision; Flask or Django for backend APIs.
- **Blockchain Tools:** Ethereum testnet or Hyperledger Fabric for implementing smart contracts.
- **Databases:** MySQL or MongoDB for storing farmer data, spare part records, and supplier details.
- **GIS Tools:** Google Maps API or OpenStreetMap for geospatial supplier recommendations.

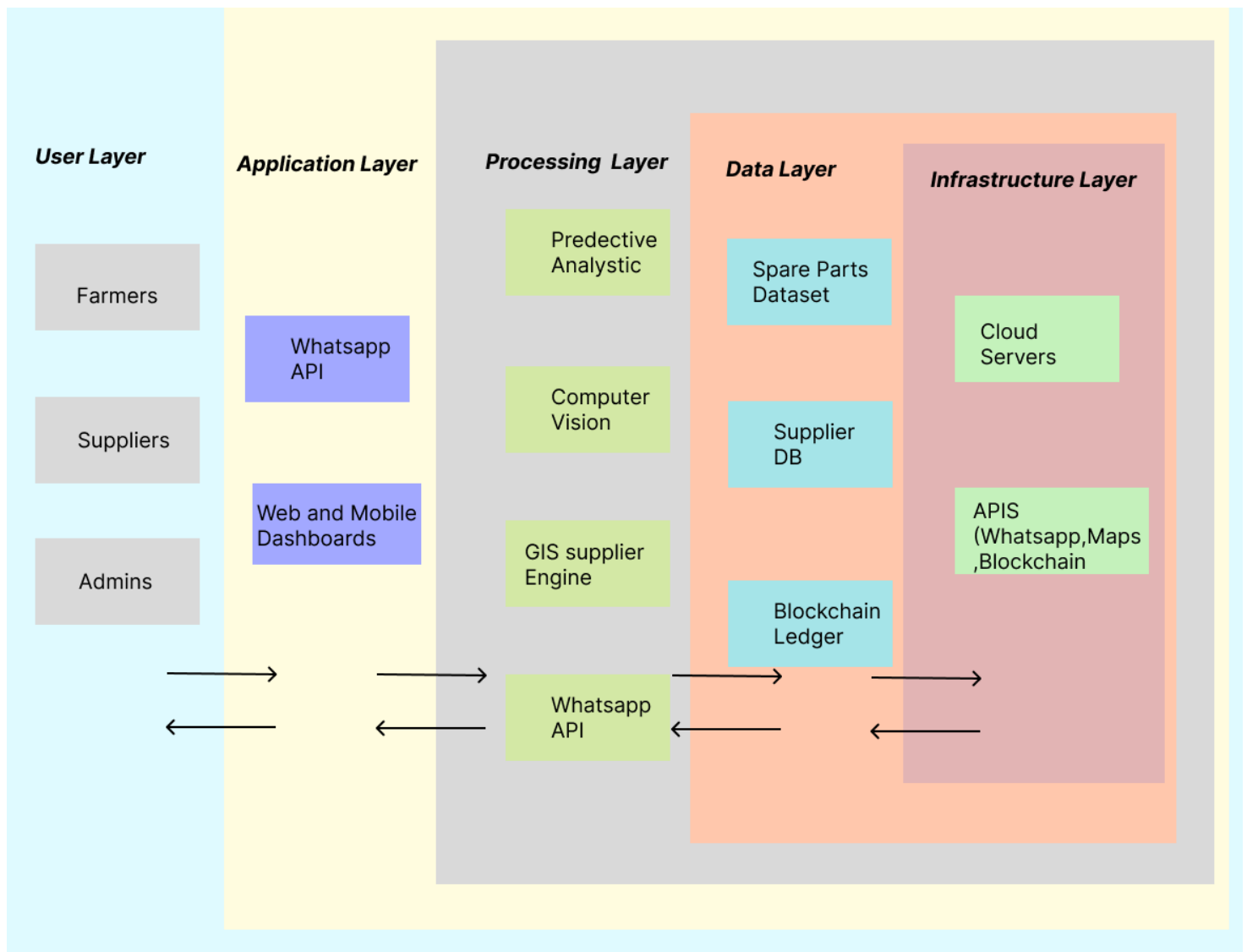


- **Development Tools:** Git for version control, Google Collab or AWS for model training, WhatsApp Business API for communication.

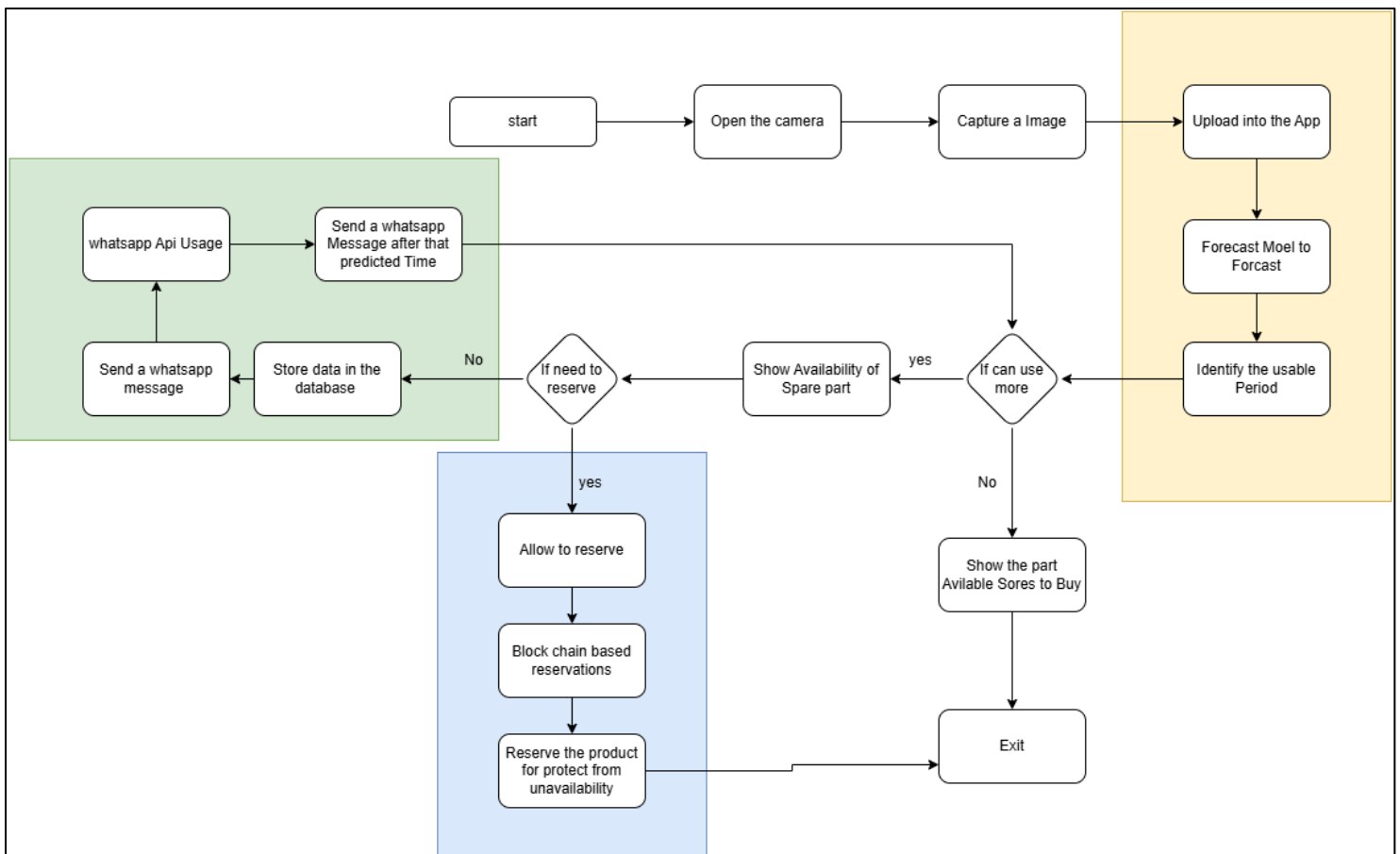
#### 4.4 User Requirements

- **Ease of Use:** Farmers must be able to interact with the system using simple WhatsApp messages without needing technical expertise.
- **Language Support:** Interfaces and notifications should be available in Sinhala and Tamil to ensure inclusivity
- **Trust and Transparency:** Farmers should be able to verify part authenticity through blockchain verification.
- **Proactive Engagement:** Farmers should receive automatic alerts before part failures occur.
- **Accessibility Across Devices:** The system should function on low-end smartphones and in areas with limited internet connectivity.

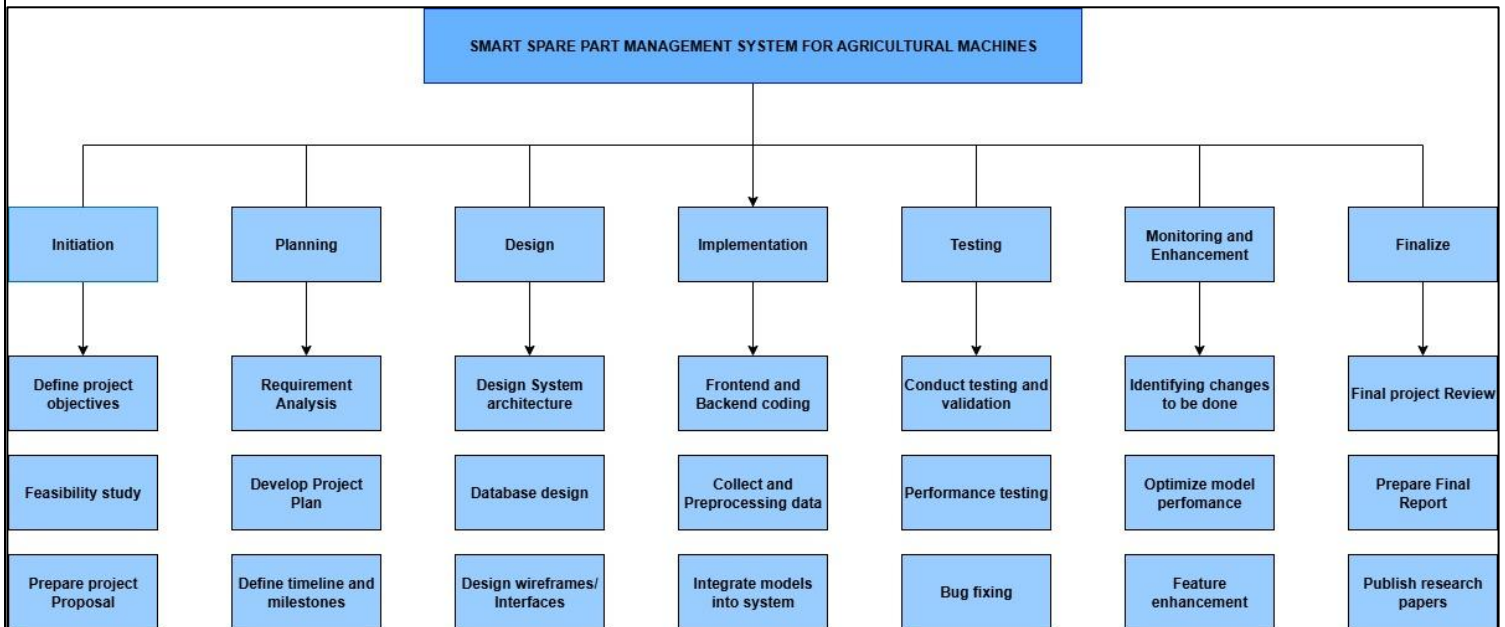
## 5. System Architecture Diagram



## 6.Component Flow Diagram



## 7. Work Break Down



## 8. Gantt Chart

|                                    | 2025 |      |      |     |     |     |     |     | 2026 |     |       |       |     |
|------------------------------------|------|------|------|-----|-----|-----|-----|-----|------|-----|-------|-------|-----|
| Task                               | May  | June | July | Aug | Sep | Oct | Nov | Dec | Jan  | Feb | March | April | May |
| Research Topic Selection           |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Topic Evaluation Form              |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Feasibility Study                  |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Requirement Gathering              |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Proposal                   |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Proposal Presentation      |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Proposal Documentation     |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Software Requirement Specification |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Software Designing                 |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Software Implementation            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Unit Testing                       |      |      |      |     |     |     |     |     |      |     |       |       |     |
| System Integration Testing         |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project Status Document            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Progress Presentation 1            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| writing Research Papers            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Testing Level 2                    |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Progress Presentation 2            |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Demo Level 1                       |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Website Assesment                  |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Project status Document 2          |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Log Book                           |      |      |      |     |     |     |     |     |      |     |       |       |     |
| Final Presentation and viva        |      |      |      |     |     |     |     |     |      |     |       |       |     |

## 9. Commercialization, Budget and Justifications

### 9.1 Market Demand

- **Growing Need for Predictive Maintenance:** With agriculture mechanization increasing, there is a high demand for reliable spare part management systems.
- **Localized Solutions:** Current predictive maintenance platforms are industrial-focused and do not address Sri Lanka's agricultural context.
- **Trust Issues:** Farmers often face counterfeit spare parts, creating a strong need for secure, transparent supply chains.

### 9.2 Commercial Viability

- **Product Offering:** The system can be deployed as a SaaS (Software as a Service) model for agricultural institutions, cooperatives, and suppliers.
- **Revenue Models:**
  - Subscription-based services for suppliers and farmer associations
  - Freemium model (basic version free, advanced features paid)

### 9.3 Competitive Advantage

- **Integration of AI, Blockchain, and GIS:** Unique combination not yet available in Sri Lankan agriculture.
- **Localized Language Support:** Sinhala and Tamil interfaces make the system farmer friendly.
- **Trust and Security:** Blockchain verification prevents counterfeit parts, building credibility in rural supply chains.
- **Scalable and Cost Effective:** Uses cloud-based infrastructure to minimize costs while serving many users.

### 9.4 Justification

- **Cost Efficiency:** Using cloud services reduces upfront hardware costs and provides flexibility.
- **Scalability:** The system can scale with user demand, ensuring reliability during peak usage.
- **Inclusivity:** Investment in WhatsApp based services ensures accessibility to farmers with low digital literacy.
- **Long Term Sustainability:** Blockchain integration provides secure, transparent operations that build trust and reduce counterfeiting risks.

## 10. Conclusion

This proposal presents the design and development of an **AI-enhanced Spare Parts Lifecycle Forecasting and User Engagement System** tailored specifically for the Sri Lankan agricultural sector. The proposed solution combines a wide range of cutting-edge technologies including **predictive analytics, computer vision, multilingual communication platforms, GIS-based supplier recommendations, and blockchain-enabled authenticity verification**. By integrating these technologies into a single mobile-based ecosystem, the project directly addresses some of the most pressing challenges faced by local farmers, such as unexpected machinery breakdowns, the circulation of counterfeit spare parts, lack of timely communication, and difficulties in accessing nearby verified suppliers.

The contribution of this research is significant because it not only focuses on **technological innovation** but also ensures **practical inclusivity**. By forecasting spare part degradation through AI-driven predictive models, farmers can be alerted well in advance about potential failures. The use of computer vision further strengthens the solution by allowing farmers to simply capture an image of a spare part and receive instant feedback on its condition. This process reduces dependency on technicians and provides a cost-effective diagnostic tool that is accessible to even small-scale farmers with minimal digital literacy. The inclusion of multilingual communication channels, particularly through WhatsApp in Sinhala and Tamil, ensures that the system is not restricted to English-speaking users and remains usable in diverse rural contexts.

Furthermore, the **blockchain-powered authenticity verification system** is a major innovation within the agricultural machinery spare parts domain. It not only guarantees the originality of purchased spare parts but also builds trust between farmers and suppliers, addressing one of the most common pain points in rural supply chains. Coupled with a **geospatial recommendation engine**, the system empowers farmers by offering accurate, location-based recommendations of the nearest verified suppliers. This reduces the time and cost associated with finding replacement parts, especially during peak harvesting or planting periods when downtime can lead to significant economic losses.

The anticipated outcomes of the system extend beyond just technical efficiency. If successfully implemented, the solution will lead to **reduced downtime, improved agricultural productivity, enhanced farmer engagement, and stronger trust in supply networks**. On a larger scale, it contributes to the modernization of Sri Lanka's agricultural ecosystem, paving the way for the adoption of **smart farming practices** and **digitally enabled rural economies**. The project also has the potential to expand into other developing nations facing similar challenges, thereby offering global scalability.

In conclusion, this proposal not only introduces an innovative and scalable solution but also emphasizes **cost-effectiveness, inclusivity, and long-term sustainability**. By leveraging advanced technologies while keeping farmer usability at the core, the system bridges the gap between high-tech predictive maintenance methods and the real-world constraints of rural communities. Future extensions of this research could involve integration with IoT sensors on

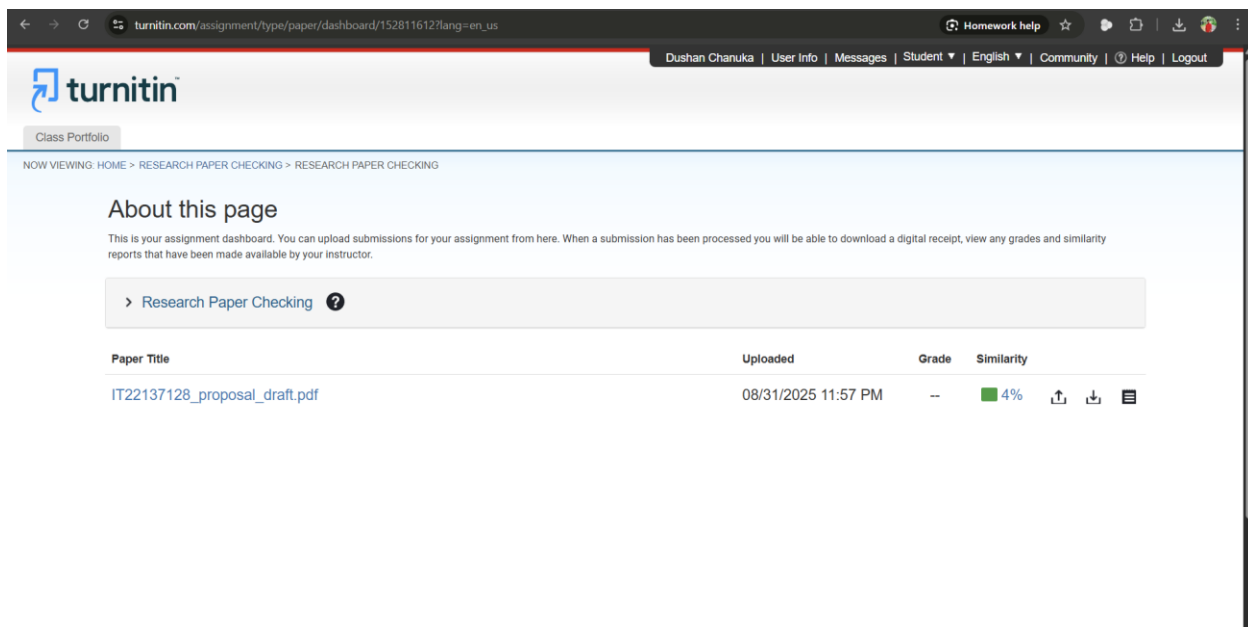
machinery for real-time monitoring, larger datasets covering multiple regions of Sri Lanka, and partnerships with government institutions for nationwide adoption. Ultimately, the system represents a meaningful step toward modernizing agriculture in Sri Lanka, strengthening food security, reducing financial losses for farmers, and improving both **economic resilience and social sustainability** in the country.

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## 12. Appendices



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